

Civil Engineering Department

MODEL EXIT EXAM

1. Select the incorrect combination regarding peculiar features of soil as an engineering material
 - a) Particulate material → made up of countless particles of a variety of shapes and sizes with little or no bonding between them
 - b) Multi-phase material → solid, liquid, gas phases coexist in a soil mass
 - c) Stress-dependent stiffness → the stiffness and strength of a given soil is fixed
 - d) Memory capacity → soil remembers the maximum stress it has been exposed to in its entire existence
2. Which of the following statement is correct regarding field identification of soils?
 - a) The presence of an appreciable quantity of organic material can usually be recognized by the dark grey to black color and odor of decaying vegetation
 - b) Since silt soils contain much smaller voids than clays and are much less permeable, the appearance of the surface of pat does not change during the shaking test
 - c) In dry strength test, a silt fragment can be broken only with a great effort, whereas a clay fragment crush easily
 - d) If a sample of moist soil can be manipulated between the palms of the hands and fingers and rolled into a long thread of about 3mm diameter, the soil then contains a significant amount of silt; whereas clay cannot be rolled into a thread of 3mm diameter without sever cracking.
3. Select the incorrect statement regarding infiltration, percolation, and seepage.
 - a) Infiltration: a process by which water enters the ground through a porous surface.
 - b) Percolation: the downward movement of water (after it has entered the ground) under the influence of gravity.
 - c) Seepage: the general movement of water through a soil mass due to hydraulic gradient.
 - d) Seepage flow is one directional only and is uniform over the entire area perpendicular to the flow.
4. Which of the following statement is not true regarding about soil improvement through compaction?
 - a) Soil mass is compacted in layers called lifts in road construction to ensure the compactive effort applied reaches all points within the soil mass.
 - b) The fundamental reason for the development of modified Proctor test is the development of larger and heavier machines to increase the amount of compaction of the soil and since it was found that the Standard Compaction test could not reproduce the densities measured in the field.
 - c) During sample preparation for compaction test, one of the best ways to make sure the

provided moisture is evenly distributed through the soil is using fresh soil sample as much as possible.

- d) During compaction in the field, it is common practice to prefer the wet side of optimum moisture content than the dry one.

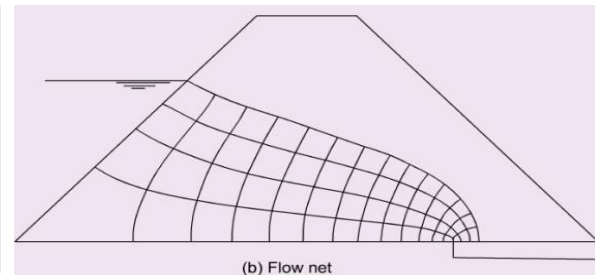
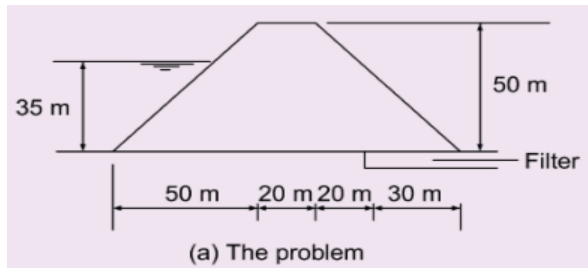
5. The cross-section of an earth dam is shown in the figure below. If the water level remains constant at 35 m, determine the seepage loss through the dam in m^3/day . The width of the dam is 300 m, and the soil is isotropic with $k = 5.8 \times 10^{-7} \text{ m/s}$.

a) 0.5

b) 4

c) 14

d) 150



6. Which of the following statements is incorrect regarding stresses within a soil mass.
- The total vertical stress acting at a point in the soil is due to the weight of everything that lies above that point including soil, water and any load applied to the soil surface.
 - The magnitude of the pore pressure at a point in the soil depends on the depth below the water table and the flow conditions.
 - Effective stress = Total stress + Pore pressure
 - Many geotechnical structures operate in a state of plane strain, i.e. one dimension of the structure is large enough for end effects to be ignored and the problem can be regarded as one of two dimensions.
7. Which of the following statements is correct regarding bulbs of pressure?
- Bulbs of pressure are constructed by plotting points of equal vertical pressure on a cross-section through the foundation
 - From a bulb of pressure one has some idea of the depth of soil affected by a foundation
 - Small foundations will act together as one large foundation resulting in overlapping of pressure bulbs unless the foundations are at a greater distance apart (c/c) than five times their width
 - All are correct
8. An infinite slope is made of sand with a friction angle of 32° and a unit weight of 17 kN/m^3 as part of the currently under-construction Modjo-Hawassa express route. The slope angle is 2.5 horizontal to 1 vertical. What are the factors of safety against slope failure i) in the “spring/ጥንታዊ” when the slope has no water, ii) for the same slope on the moon respectively. (The acceleration due to gravity on the surface of the Moon is about 1.625 m/s^2 i.e. 16.6% of that on Earth’s surface. Assume that there is no water on the moon)

- a) 2.5 & 0.4 b) 2.5 & 1 c) 1.56 & 1.56 d) 1.56 & 0.26

9. Select the incorrect combination of settlement component and corresponding cause among the following choices.

- a) Immediate settlement (Elastic Settlement) → caused by the elastic deformation of dry soil and of moist and saturated soils without any change in the moisture content.
- b) Primary consolidation settlement (Consolidation) → result of a volume change in saturated cohesive soil because of expulsion of the water that occupies the void spaces.
- c) Secondary consolidation settlement (Creep) → induced because of primary consolidation
- d) Secondary consolidation settlement (Creep) → observed in saturated cohesive soils and is the result of the plastic adjustment of soil fabrics.

10. Select the incorrect statement from the following.

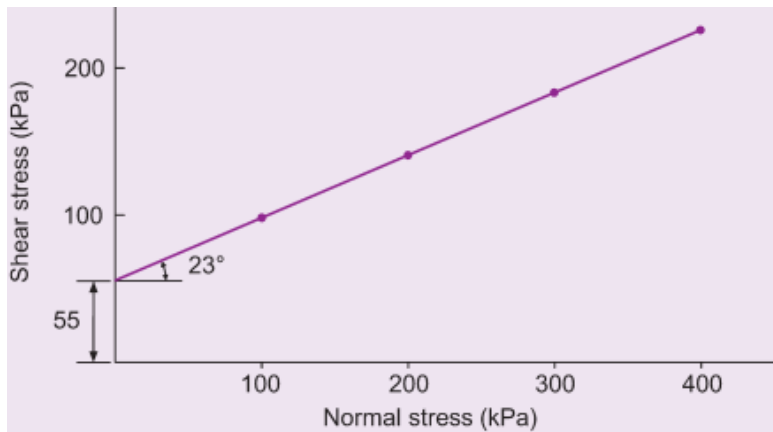
- a) An elastic analysis of an isotropic material involves only two constants—Young's modulus and Poisson's ratio—and thus if we assume that soils are isotropic elastic materials, then we have a powerful, but simple, analytical tool to predict a soil's response under loading.
- b) Both consolidation and compaction processes result in reduction in volume, but the latter involves sustained loading as opposed to the instant load application in the former.
- c) The major reason for geotechnical engineers opt not to generally use direct shear test for determination of shear strength parameters of fine grained soils is because the excess pore pressure developed in the specimen will offset the results significantly.
- d) If soil mass is semi-infinite, homogeneous, dry, cohesionless AND the face of the wall in contact with the backfill is vertical and smooth, Rankine's earth pressure theory and Coulomb's earth pressure theory give equal results

11. Which of the following statement is not true in relation to soil failure criterion?

- a) Tresca failure criterion is relevant for loading on cohesive soils (clays and fine silts) when the condition may be assessed as undrained.
- b) Mohr-Coulomb failure criterion is used in drained conditions where a change in stress also leads to a change in effective stresses.
- c) Tresca failure criterion is defined by the undrained shear strength, S_u , being equal to the maximum shear stress at failure.
- d) Mohr-Coulomb failure criterion states that a material fails because of its either maximum normal or shear stress alone.

12. Drained shear box tests were carried out on a series of soil samples to establish strength envelope and the best straight line through the points. Select the corresponding shear strength

parameters.



- a) $C_u = 55 \text{ kPa}$
- b) $C' = 55 \text{ kPa}$
- c) $\phi_u = 23^\circ$
- d) $\phi' = 46^\circ$

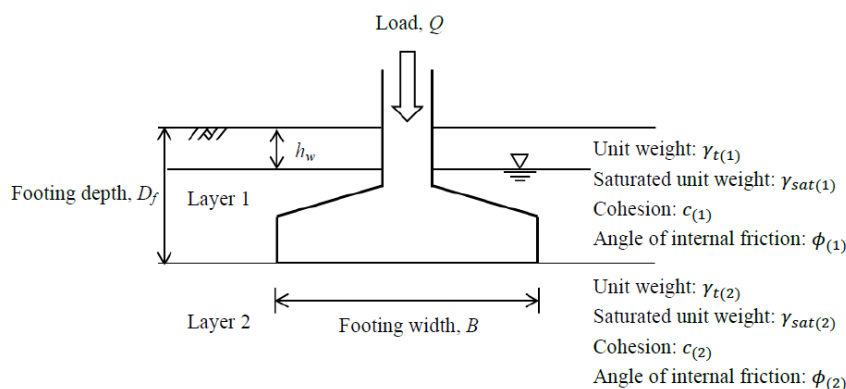
13. The results of a drained triaxial test on normally consolidated clay ($C=0$) are $\sigma'_3 = 276 \text{ kPa}$ and $(\Delta\sigma_d)_f = 276 \text{ kPa}$. Select the incorrect value among the pertinent parameters.

- a) $\sigma_1 = 552 \text{ kPa}$
- b) $\phi = 49.45^\circ$
- c) $\sigma_f = 368 \text{ kPa}$
- d) $\tau_f = 130 \text{ kPa}$

14. Which one of the following statements is incorrect regarding soil modulus?

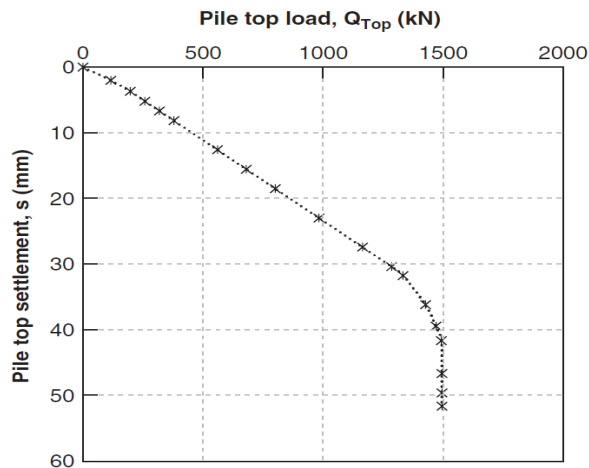
- a) The shear modulus, G , relates to strain response of a body to shear or torsional stress i.e. accounts for change of shape without change of volume.
- b) The bulk modulus, K , describes the strain response of a body to hydrostatic stress i.e. accounts for change in volume without change of shape.
- c) Young's modulus, E , accounts for both change of shape without change of volume AND change in volume without change of shape; $E=G+K$
- d) Modulus of subgrade reaction is not a true soil property since it depends on the size of the loaded area.

15. Select the correct modified form of the ultimate bearing equation ($q_u = cN_c + \gamma D_f N_q + 0.5\gamma B N_\gamma$) taking into account the prevailing conditions shown in the figure below.



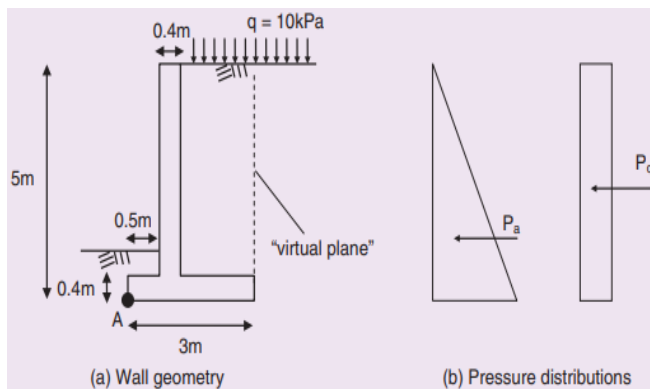
- a) $q_u = c_{(1)}N_c + [(\gamma_{t(1)} * h_w) + (\gamma_{sat(1)} - \gamma_w)(D_f - h_w)]N_q + 0.5(\gamma_{sat(1)} - \gamma_w)BN_\gamma$
- b) $q_u = c_{(2)}N_c + [(\gamma_{t(2)} * h_w) + (\gamma_{sat(2)} - \gamma_w)(D_f - h_w)]N_q + 0.5(\gamma_{sat(2)} - \gamma_w)BN_\gamma$
- c) $q_u = c_{(1)}N_c + [(\gamma_{t(2)} * h_w) + (\gamma_{sat(2)} - \gamma_w)(D_f - h_w)]N_q + 0.5(\gamma_{sat(1)} - \gamma_w)BN_\gamma$
- d) $q_u = c_{(2)}N_c + [(\gamma_{t(1)} * h_w) + (\gamma_{sat(1)} - \gamma_w)(D_f - h_w)]N_q + 0.5(\gamma_{sat(2)} - \gamma_w)BN_\gamma$

16. Which of the following statements is true regarding soil investigation via boreholes?
- The number and spacing of bore holes depend upon the extent of the site and the nature of structures coming on it
 - Boreholes should generally extend to a depth where the stress increase from the superstructural load has almost negligible effect on the underlying soil
 - Borehole location should be distributed in such a way that foundation soil layering can be determined across the location
 - All are true statements.
17. A silty sand of density index, $I_D = 60\%$ was subjected to standard penetration tests at a depth of 3m. Groundwater level occurred at a depth of 1.5 m below the surface of the soil which was saturated throughout and had a unit weight of 19.3 kN/m^3 . The average N count was 15. During calibration of the test equipment, the energy ratio has been determined to be 74 %. What is the $(N_1)_{60}$ value for the soil?
- 15
 - 19
 - 26
 - 60
18. What is the smallest isolated square footing size to carry a column load of 750 kN? The depth of the footing is 3 m and the unit weight of the sandy soil ($\phi = 29^\circ$) is 18 kN/m^3 . The corrected bearing capacity factors for $\phi = 29^\circ$ are $N_c = 45$, $N_q = 20$, $N_\gamma = 13$. Minimum factor of safety based on ultimate bearing capacity = 3.0
- 1 m
 - 1.5 m
 - 2 m
 - 2.5 m
19. Consider the following field tests and rank them in decreasing order of their suitability in pile foundation design.
- | | |
|-------------------------------|------------|
| [1] Standard Penetration Test | a) 4-1-3-2 |
| [2] Vane Shear Test | b) 4-3-1-2 |
| [3] Cone Penetration Test | c) 3-4-2-1 |
| [4] Pile Loading Test | d) 4-2-3-1 |
20. A concrete pile (diameter $D = 0.5 \text{ m}$, length $L = 12 \text{ m}$) is embedded in a deep clay (cohesion = 110 kPa) deposit. The bearing capacity factors are $N_c = 9$, $N_q = 1$, $N_\gamma = 0$. The unit friction along the sides of the pile is 65 kPa . What is the ultimate bearing capacity of the pile?
- 1100kN
 - 650 kN
 - 1420 kN
 - 1225 kN
21. Which one of the following is not correct about a pile group carrying vertical loads?
- The load capacity of a pile group may not necessarily be the load capacity of a single pile multiplied by the number of piles.
 - Piles driven into a loose, coarse-grained soil tend to densify the soil around them
 - Piles might have global efficiency higher than one
 - None of the above
22. The following plot is obtained from a pile load test. Estimate the ultimate pile capacity.



- a) 1250 kN
- b) 1500 kN
- c) 30 mm
- d) 50 mm

23. The proposed design of a cantilever retaining wall is shown in figure below. The unit weight of the concrete is 25 kN/m^3 and the soil has weight density 18 kN/m^3 . The soil peak strength parameters are $\phi' = 38^\circ$, $c' = 0$ and the design bearing resistance of the soil beneath the wall has been calculated to be 250 kPa . The retained soil carries a uniform surcharge of intensity 10 kPa . Ignore any passive resistance from the soil in front of the wall. Assume coefficient of friction between base of wall and soil to equal $\tan \phi'_{\text{peak}}$. Which of the following statements is correct?



- a) Factor of safety against sliding, $F_s = 3.2$
- b) Factor of safety against overturning, $F_o = 3.8$
- c) Factor of safety against bearing capacity failure, $F_b = 1.85$
- d) All are correct

24. Of the following road safety measures, which one has produced the biggest road safety impact on a global scale?

- a. Road safety education
- b. Using advertisements to modify driver behavior
- c. Monitoring driver alcohol levels
- d. Design of safe systems, i.e., vehicles, infrastructure and post-crash care

25. The level of service of a highway segment is "F" whenever:

- a. $v/c > 1$
- b. $v/c < 1$
- c. $v = c$
- d. a & c

26. The number of vehicles that pass through a transverse line of road at a given time in a specified direction is called _____

- a. traffic studies
- b. traffic flow
- c. traffic origin
- d. traffic destination

27. Link travel time function relates travel time and

- a. link volume
- b. link cost
- c. level of service
- d. none of the above

28. Which one is the most widely used traffic count procedure for the design of roadways?

- a. A One month 24 hrs count
- b. A 7 day 24 hrs count
- c. A 7 day count of which two days are for 24 hrs and 5 days for 12 hrs
- d. A 7 day count of which two days are for 24 hrs and 5 days for 16 hrs

29. In highway design, curves have been divided into two broader classes. They are:

- a. Reverse curve vs simple curves
- b. Crust curves vs spiral curves
- c. Horizontal vs vertical curves
- d. Compound curves vs sag curves

30. Raising of the outer edge of horizontal curves can be called

- a. Widening
- b. simple curve
- c. super elevation
- d. all of the above

31. Factors that affect the design of horizontal curves include:

- a. Object height
- b. Design speed of the vehicle
- c. Allowable friction
- d. b and c

32. A heavy bus navigates with a speed of 40 km per hour, the brake reaction time, in normal cases,

- a. 7 seconds
- b. 0.5 seconds
- c. 2 seconds
- d. 2.5 second

33. Type of At-grade intersections include:

- a. priority intersections
- b. roundabout intersections

- c. signalized intersections
- d. all of the above

34. Addis Ababa –Adama road can be classified as

- a. Access road
- b. Collector road
- c. Serves as mobility road
- d. Link road

35. The highway facilities are design for

- a. Annual Average Daily Traffic
- b. Peak hour volume
- c. The 30th Highest hourly volume
- d. Weekly average Daily traffic

36. Bottom-most component of a flexible pavement, is

- a. Sub-grade
- b. Sub-base
- c. Base
- d. Base course

37. Which of the following is considered to be the highest quality construction in the group of black top pavements?

- a. Mastic asphalt
- b. Sheet asphalt
- c. Bituminous carpet
- d. Bituminous concrete

38. Los Angeles testing machine is used to conduct

- a. Abrasion test
- b. Impact test
- c. Attrition test
- d. Crushing strength test

39. When the bituminous surfacing is done on already existing black top road or over existing cement concrete road, the type of treatment given is

- a. Seal coat
- b. Tack coat
- c. Prime coat
- d. Spray of emulsion

40. The function of an expansion joint in rigid pavements is to

- a. Relieve warping stresses
- b. Relieve shrinkage stresses

- c. Resist stresses due to expansion
- d. Allow free expansion

41. Penetration test on bitumen is used for determining its

- a. Grade
- b. Viscosity
- c. Ductility
- d. Temperature susceptibility

42. The fundamental factor in the selection of pavement type is

- a. Climatic condition
- b. Type and intensity of traffic
- c. Sub-grade soil and drainage conditions
- d. Availability of funds for the construction project

43. Pavement is said to be flexible if it contains

- a. Water bound macadam surface
- b. Stabilized soil base constructed of lime cement or tar
- c. Bitumen-bound stone layer of varying aggregates
- d. All the above

44. Design of flexible pavements is based on

- a. Mathematical analysis
- b. Empirical formula
- c. A compromise of pure theory and pure empirical formula
- d. None of these

45. The binder normally used in flexible pavement construction is

- a. Cement
- b. Lime
- c. Bitumen
- d. None of the above

46. The purely residential zone has _____

- a. No attraction trips
- b. No production trips
- c. Both attraction and production trips
- d. None

47. From the data collected shown below, what will be the total ESALs of the vehicles shown, you can assume one vehicle in each category.

No.	Vehicle Category	Design vehicle in each category	Axle Load (kN)			
			Single Axle	Tandem Axle	Tandem Axle	Tandem Axle
1	Bus	1	50	-	102	-
2	Medium Truck	1	55	-	124	-
3	Heavy Trucks	1	56	126	100	-
4	Heavy Trucks trailer	1	58	150	116	182

- a. 5.62
- b. 7.22
- c. 4.42
- d. 6.18

48. A soil sample test for CBR gave the following result

Penetration (mm)	2.54	5.08
Load (MPa)	0.69	1.13

If the standard load at 2.54mm and 5.08mm penetration are 6.9MPa and 13MPa respectively, what is the CBR value of the soil sample?

- a. 8%
- b. 11%
- c. 10%
- d. 12%

49. Which of the following is a destructive test for materials?

- a. Concrete cube compressive test
- b. Tensile strength test of rebar
- c. Unit weight measurement of concrete
- d. All of the above

50. Which of the following cannot be shown on site plan drawing?

- a. Survey beacon
- b. Site orientation
- c. Utilities
- d. None

51. Which one of the following is part of a stair?

- a. Stringer
- b. Landing
- c. Baluster
- d. All of the above

52. The practice of planning, designing, constructing, and operating buildings is most usually a collective effort of different groups of professionals and trades. Which one of the following is not a responsibility of a consultant?
- a. Issue provisional and final acceptance certification
 - b. Prepare contract documents
 - c. Resolve contractual disputes
 - d. Provide post occupancy repair & maintenance if required
53. Which one of the followings is not the advantage of quantity surveying:
- a. Certification of contractor payments
 - b. Handling variations works at project site
 - c. As procedures for testing and acceptance of construction works
 - d. Assist in the accurate preparation of tenders
54. The advantage of prefabricated building structural system as compared to cast in-situ type is,
- a. It requires less formwork
 - b. More time efficient
 - c. Very heavy to transport
 - d. a and b
55. Failure of formwork is expensive and embarrassing for everyone involved in the project. Which of the following cannot be a reason for formwork failure in building construction site?
- a. Improper or inadequate shoring
 - b. Lack of control of placement of concrete
 - c. Negligence of workers or supervisors
 - d. None
56. The process each manager follows during life cycle of a project is known as: _____
- a. Project management
 - b. Manger life cycle
 - c. Project Management life cycle
 - d. All of the mentioned
57. A document prepared for the customer during project initiation and planning that describes what the project will deliver and outlines generally at a high level all work required to complete the project is the:
- a. Information Systems Plan
 - b. Statement of Work
 - c. Mission Statement
 - d. Baseline Project Plan
58. In which type of contract does the contractor offers to do the whole work as shown in drawings and described by specifications, for a total stipulated sum of money?

- a. Item rate contract
- b. Lump sum contract**
- c. Cost plus fixed fee contract
- d. Cost plus percentage of cost contract

59. What is the first step in construction planning?

- a. Establish the objectives and scope.**
- b. Determine the budget.
- c. Select the team organizational model.
- d. Inspect the deliverables

60. Which of the following test methods is not appropriate for testing workability of concrete?

- a. Slump test
- b. Soundness test**
- c. Compaction factor test
- d. Vebe time test

61. When a consultant prepares specifications and construction drawings that contain a misrepresentation the consultant may be found

- a. Responsible for subsequent additional costs that might be forthcoming.**
- b. Protected by indemnification clauses in the contract between the owner and the contractor.
- c. Protected by indemnification clauses in the contract between the owner and the consultant.
- d. Responsible for providing subsequent site instructions to the owner.

62. Construction cost estimation is the process of determining probable cost of a project made before commencement of a project. The estimator should consider the different factors that will affect the project and these are

- a. Nature of subsurface condition
- b. The method of bid evaluation
- c. Location of the project site
- d. All of the above**

63. In this type of dispute resolution mechanism, a neutral third party which preferably has experience and knowledge about the issue gives specific suggestions and advises the disputing parties on how to reach an agreement.

- a. Mediation
- b. Adjudication
- c. Conciliation**
- d. Arbitration

64. During which phase must project teams address important considerations for managing

information (and often end up updating business processes through improved communications)?

- a) Closing
- b) Execution
- c) Planning
- d) Monitoring and controlling

65. What is Quality Assurance?

- a) It is the measurement of degree to which a product satisfies the need
- b) Process of identifying defects
- c) Any systematic process used to ensure quality in the process
- d) It is a corrective tool

66. Portion of dam in contact with ground at downstream side is

- A. Crest
- B. Toe
- C. Heel
- D. Foot

67. _____ is the arrangement made near top of dam for passage of excess water from the reservoir.

- A. Sluiceway
- B. Spillway
- C. Gallery
- D. Abutment

68. The storage of water below the bottom of the lowest sluiceway in a reservoir is:

- A. Dead storage
- B. Surcharge storage
- C. Live storage
- D. Useful storage

69. If the area of drainage basin is double, other factors remaining the same the peak discharge according to Rational Method will be approximately

- A. 2 times the original discharge
- B. Half of original discharge
- C. The same as the original discharge
- D. 1.5 times the original discharge

70. Seepage endangers the stability of an earth dam built on pervious foundation because of

pipings which depends on

- A. Height of the dam
- B. Quantity of seepage flow
- C. Value of the exit gradient
- D. Total storage capacity of the reservoir

71. When the reservoir is full the maximum compressive stress in a gravity dam is produced

- A. At the heel
- B. At the toe
- C. Within the middle third of the base
- D. At the center of the base

72. Hydrograph is a plot of

- A. Rainfall intensity against time
- B. Cumulative rainfall against time
- C. Discharge against time
- D. Cumulative runoff against time

73. What are the principles governing the conceptual design of the Earthquake resistant building?

- a) Structural Simplicity, uniformity, symmetry and redundancy
- b) Energy dissipation capacity
- c) bi-directional resistance and stiffness
- d) All of the above

74. Which one is true about concrete cover.

- a) It will enhance the durability of the structure.
- b) Do not contribute to the bond strength between concrete and steel
- c) It is used to protect the rebar from fire, soil and chemical
- d) all of the above

75. Concrete cover is provided in order to ensure the following except _____.

- a. Safe transmission of bond forces
- b. Protection of the steel against corrosion
- c. To provide flexural resistance
- d. Adequate fire resistance

76. Which one is **NOT TRUE** for the dynamic load allowance?

- a) It is an increment to be applied to the static wheel load to account for wheel load impact from moving vehicles.
- b) Hammering effect is the dynamic response of the wheel assembly
- c) Dynamic response of the bridge as a whole
- d) None

77. Which one of the following is **Not TRUE** in locating a bridge crossing?

- a) The channel in the reach should be well defined
- b) The crossing site should be as narrow as possible
- c) The crossing site should have firm high banks, which are fairly unreadable.
- d) None of the above

78. Which one is correct about Piers?

- a) It should effectively transfer loads from Superstructure to the foundation without failure.
- b) Piers should be located in such a manner that they can provide the required lineal waterway and navigational clearance.
- c) Piers should also be located to make the best use of the foundation conditions available.
- d) All of the above

79. Criteria for bridge selection include:

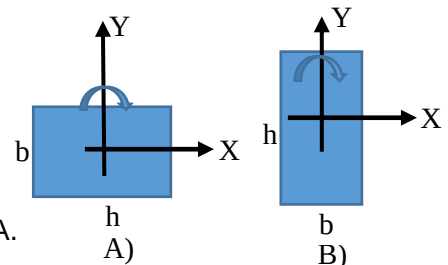
- a) Geometric Condition of the Site
- b) Functional Requirements
- c) Economy
- d) All of the above

80. Given two reinforced concrete section A and B with the same section; concrete and steel. If steel bar is provided only at tension zone and section A is provided with larger amount than section B,

- a) A is stronger and more ductile than B
- b) A is stronger but less ductile than B
- c) B is stronger and more ductile than A
- d) Both A and B have the same strength and ductility

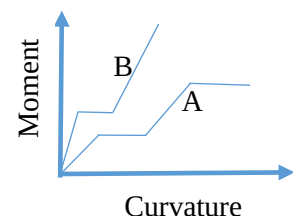
81. Which one is true about the reinforced concrete column with the same cross sections A and B with different orientation given below if h is greater than b ?

- a) Column A has more resistance than B
- b) Column B has less resistance than B
- c) They have the same resistance
- d) B has greater moment of inertia about y-axis than A.



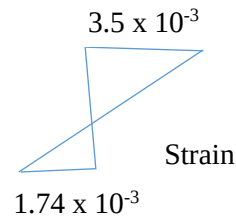
82. Which one is true about the moment curvature of two different reinforced concrete section given below?

- a) Both sections have the same ductility but different strength
- b) They have the same strength but different ductility
- c) A is stronger than B
- d) A is more ductile than B



83. Which one is true about the strain profile diagram of a reinforced concrete rectangular cross section of steel grade S-400 given below?

- a) It is tension section
- b) Compression section
- c) **Balanced section**
- d) All



84. How can you make sure that the design reinforce concrete section is ductile section.

- a) Using concrete grade
- b) Amount of reinforcement
- c) Detailing
- d) **All of the above**

85. Which one is true about doubly reinforced section?

- a) A section provided with reinforcement bar only in tension zone
- b) **It increases the ductility of the section**
- c) It does not have any relation with the ductility of the section
- d) None of the above

86. Where does the first plastic hinge form in propped cantilever reinforced concrete rectangular ductile section beam which it has symmetric reinforcement both at support and span and subjected to uniform distributed load until to fail?

- a) At mid span
- b) At pin end support
- c) **At fixed support**
- d) No plastic hinge form at any point

87. The minimum area of reinforcement for DCM design of the column section should not be less than

- a) **1%**
- b) 3%
- c) 1.5%
- d) 4%

88. The thickened part of a flat slab over its supporting column, technically known as

- a) Capital
- b) **Drop panel**
- c) Column head
- d) None of the above

89. If the effective length of reinforced concrete square column of 400mm is 5.5m, its slenderness ratio is

- a) 60

b) 65.57

c) 47.63

d) 45

90. Which of the following is/are among the advantage of reinforced concrete structures

a) Monolithic nature

b) Molding in any desire shape

c) Less maintenance cost

d) All the above

91. In reinforced concrete section

a) Steel bar is provided only at tension zone

b) Steel bar is not provided in compression zone

c) Steel bar can be placed both on tension and compression zone

d) None of these

92. The amount of secondary reinforcement in one way slab is

a) The same amount to that of the main reinforcement

b) 10% of the main reinforcement

c) Greater or equal to 20% of the main reinforcement

d) None

93. In reinforced concrete beam supported slab

a) In one way slab, more than 90% of the load is transfer in shorter direction

b) The deflection of on way slab takes bowel like shape

c) The deflection of two-way slab takes bowel shape

d) a and c

94. For the determination of the action effects in the design of building structures, linear analysis may be carried out by assuming;

a) Equivalent cracked sections with reduced stiffness

b) linear stress- strain relationships

c) mean value of the modulus of Elasticity

d) All of the above

95. Which one is incorrect in plastic analysis for beams, frames and slabs?

a) Columns should be checked for the maximum plastic moments which can be transmitted by connecting members.

b) For connections to flat slabs this moment should be included in the punching shear calculation.

c) When plastic analysis of slabs is carried out account should be taken of any non-uniform reinforcement, corner tie down forces, and torsion at free edges.

d) None of the above

96. Which one of the following is correct about steel?

- a) Homogenous and elastic material
- b) Brittle material
- c) Has good resistance to fire
- d) Less maintenance cost

97. For a trial section of the same area, which cross section is stiffer:

- a) T-section
- b) Channel section
- c) I-section
- d) Angle or L-section

98. Which of the following is not correct about design of steel structures.

- a) Theoretically, tension member is the most efficient structural element
- b) Buckling can not be considered for compression members
- c) I-section can be used for design of compression members
- d) None of the above

99. In design of steel structures, why connections are required?

- a) They are used if a change in direction occurs
- b) To form built-up members
- c) To connect members together in trusses and lattice girders
- d) All of the above

100. Which case is to be considered to discount the possibility of lateral torsional buckling in steel beams:

- a) Bending takes place only about the minor axis
- b) If the member is restrained throughout its length by adequate bracing
- c) The lateral slenderness parameter is above the limit (≤ 0.4)
- d) All of the above